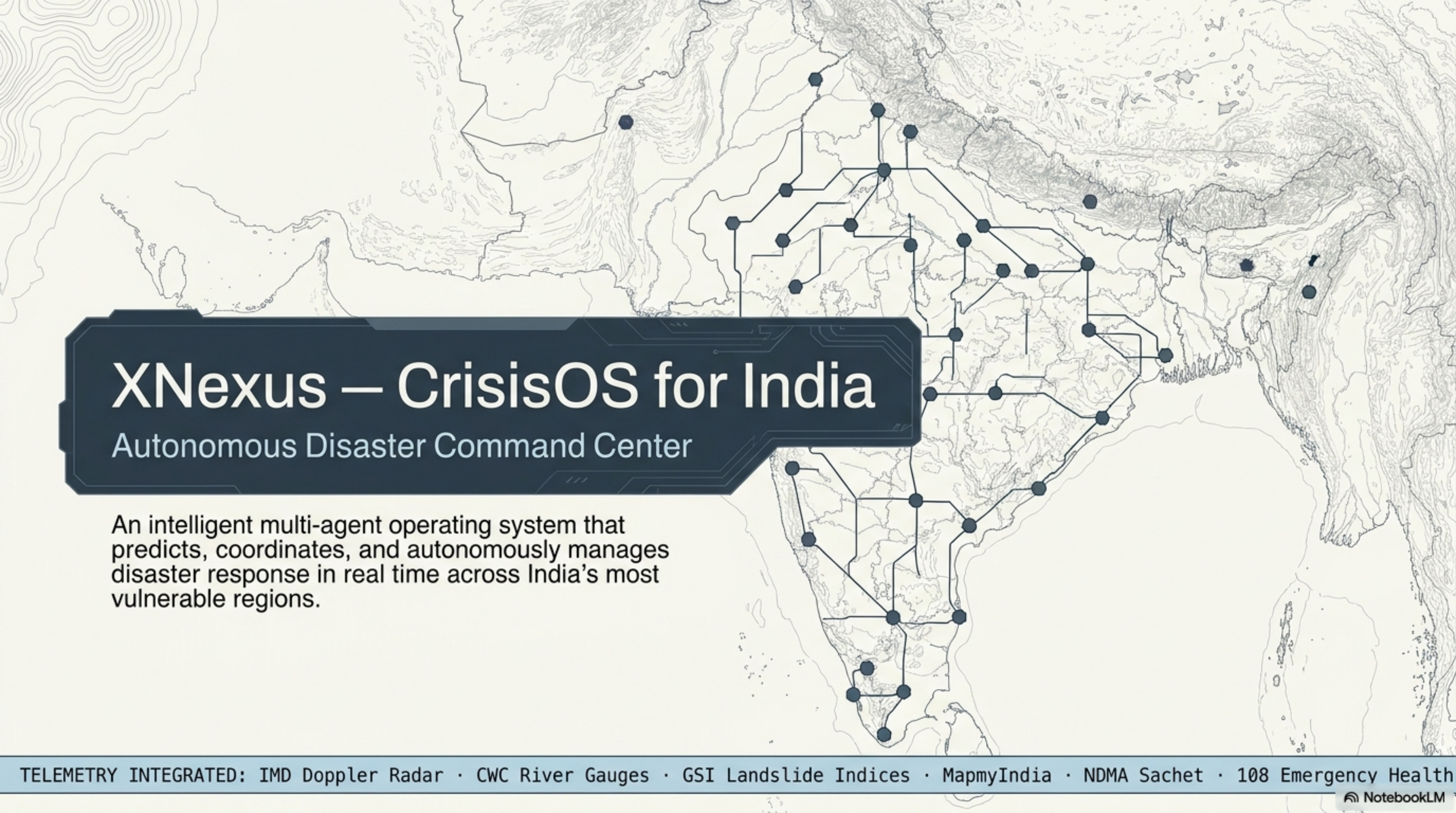




XNexus — CrisisOS for India

Autonomous Disaster Command Center

An intelligent multi-agent operating system that predicts, coordinates, and autonomously manages disaster response in real time across India's most vulnerable regions.

PEOPLE ARE DYING BECAUSE AGENCIES CANNOT COORDINATE

India possesses **world-class sensors** (IMD, CWC, GSI), but they operate in absolute **data silos**. Manual phone chains create **fatal delays**.



Diagnostic Matrix: The Lethal Cost of Unconnected Intelligence

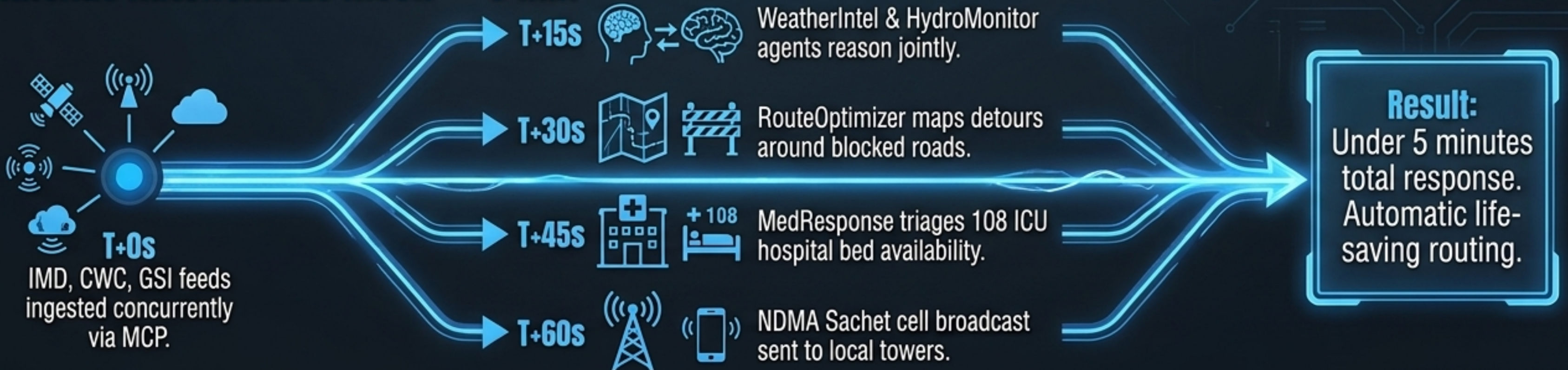
	The Siloed Failure (Historical)	The XNexus Intervention (Autonomous Resolution)
Wayanad Landslides, July 2024 - 400+ Dead	Landslides struck at 2:00 AM. CWC upstream gauges, GSI soil moisture, and IMD rain alerts were never cross-analyzed.	GeoRisk & HydroMonitor agents cross-reference soil saturation with rain rates, preemptively triggering NDMA alerts before the landslide occurs.
Kerala Floods, Aug 2018 - 483 Dead	35 dams opened simultaneously without downstream modeling. Rescue fleets arrived 6+ hours late.	CWC telemetry feeds directly into PostGIS, auto-calculating downstream inundation and routing fleets via MapmyIndia.
Cyclone Biparjoy, 2023	Standard siren broadcasts lacked evacuation route pathfinding, leading to fatal coastal traffic bottlenecks.	RouteOptimizer agent forces localized, language-specific SMS detours directly to cell towers via NDMA Sachet.

The Core Friction: 100-Minute Delay vs. The 5-Minute Mesh

Manual Phone Chain - 100+ Min Delay



XNexus Autonomous Mesh - < 5 Min



Enter **XNexus**: The Next Evolution of Disaster Command

We are replacing the manual 100-minute phone-call chain with an autonomous multi-agent AI operating system.

XNexus is not a dashboard.

Dashboards require humans to read them, interpret them, and make phone calls.

XNexus is an execution engine that predicts, coordinates, and physically dispatches disaster response without human bottlenecks.

Multi-Agent Architecture: The 9-Agent Neural Mesh



Statutory Integration: Live Indian Telemetry Data Feeds



IMD

Doppler Weather Radar. Real-time reflectivity (Z factor) and cloud azimuth density mapping.



CWC

River Gauge Network. Scrapes station water levels, discharge cusecs, and statutory danger marks.



GSI

Landslide Susceptibility. Evaluates soil saturation, shear strength, and hill slope stability.



MapmyIndia

Detour Engine. Autonomous pathfinding rerouting rescue fleets away from NHAI landslides.



NDMA Sachet

Cell Broadcast. Multilingual cell-tower SMS warnings forced to phones in hazard polygons.

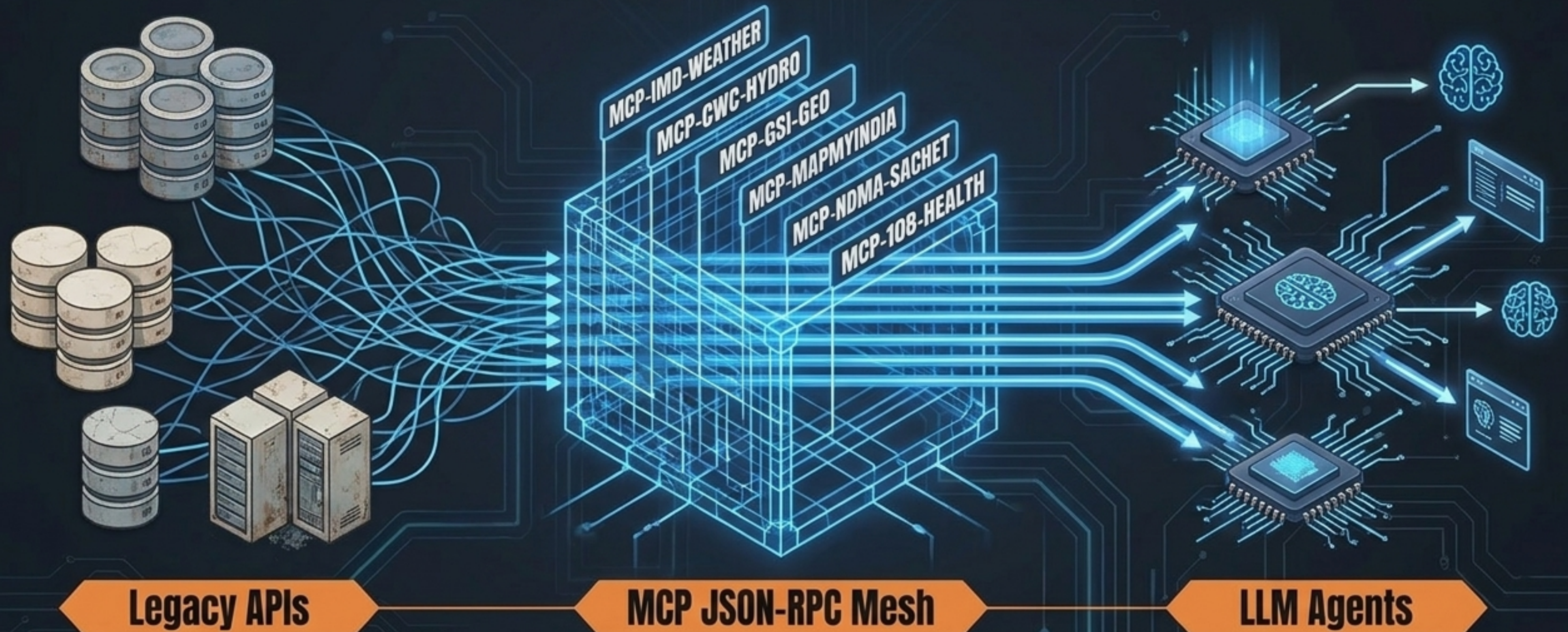


108 Health

Emergency Triage. Real-time ICU/trauma bed availability matrix and ambulance dispatch.

Architectural Foundation: Model Context Protocol (MCP) Mesh

How do we connect modern AI to legacy government APIs? We wrap every statutory endpoint into an open-standard MCP JSON-RPC tool server.



Tech Proof 01: FastMCP Ingestion Engine (CWC Telemetry)

ideation/mcp_cwc_server.py

```
from fastmcp import FastMCP
mcp = FastMCP("CWC-Hydrology-Server")
```

Exposes legacy CWC gauge data as a native LLM tool.

```
@mcp.tool()
async def get_river_level(station_id: str) -> dict:
    return CWC_DB.get(station_id)
```

Calculates exact flood margins against statutory danger marks in real-time.

```
@mcp.tool()
async def check_danger_breach(station_id: str) -> dict:
    data = await get_river_level(station_id)
    current, danger = data["current"], data["danger"]
    return {
        "status": "CRITICAL" if current >= danger else "NORMAL",
        "breach_meters": current - danger
    }
```

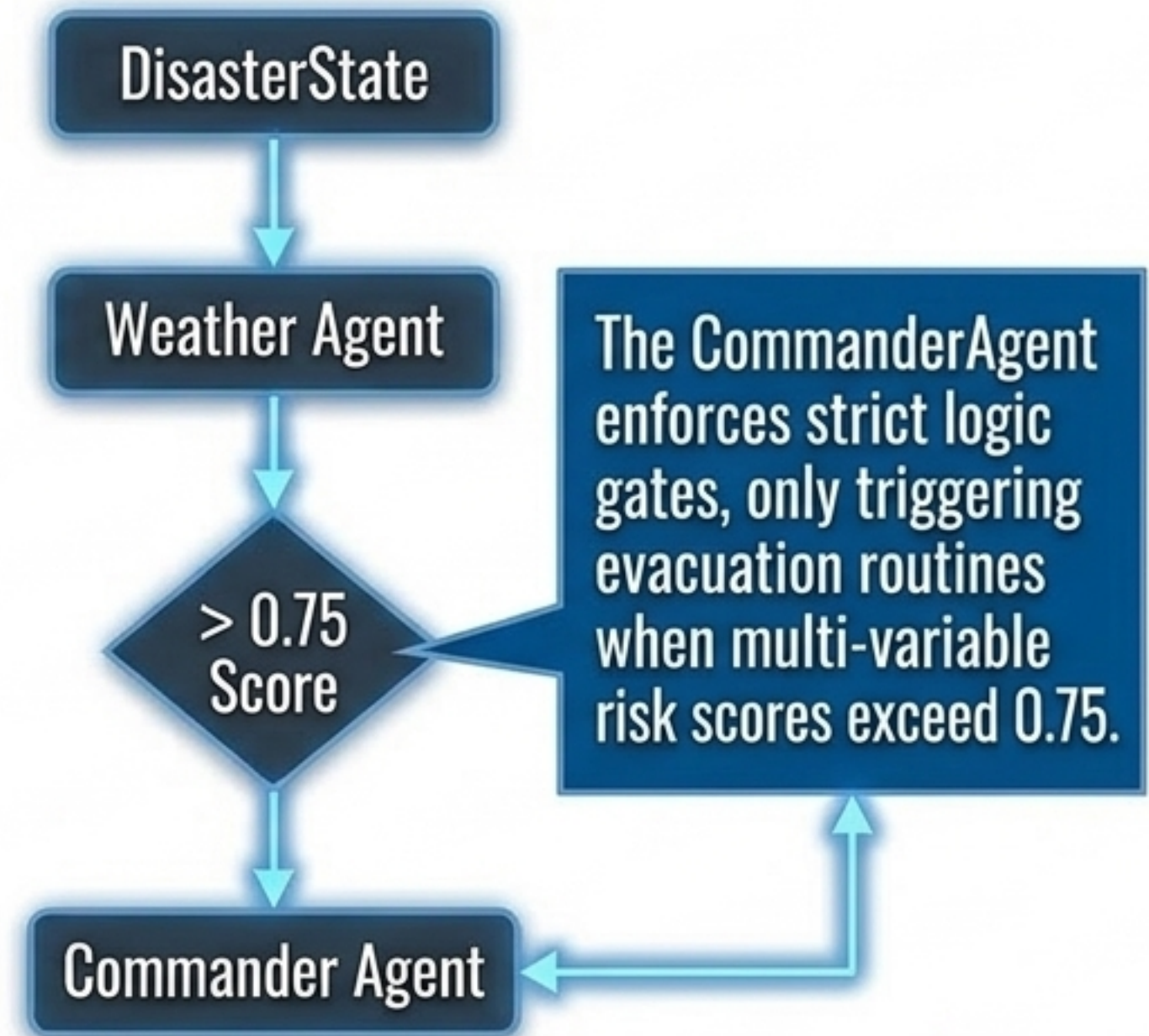
Tech Proof 02: LangGraph Multi-Agent Orchestration

ideation/agent_orchestrator.py

```
class DisasterState(TypedDict):
    incident_active: bool
    landslide_risk_score: float
    recommended_action: str
    evacuation_routes: List[str]

def commander_agent(state: DisasterState):
    if state["landslide_risk_score"] > 0.75:
        return {
            "recommended_action": "EVACUATE Sector B-4",
            "evacuation_routes": ["Route 3 East Bypass"],
            "incident_active": True
        }

workflow = StateGraph(DisasterState)
workflow.add_node("Weather", weather_agent)
workflow.add_node("Commander", commander_agent)
workflow.add_edge("Weather", "Commander")
```



Algorithmic Foundations: Deterministic Risk Modeling

Compute Engine

$$R = (Z / a) ** (1/b)$$



Z (Reflectivity)

a, b (Empirical Constants)

R (Rainfall Rate)

Translates raw Doppler radar reflectivity (Z factor ) into physical monsoon precipitation rates.

Compute Engine

$$\text{Risk} = w1 * \text{Rain} + w2 * \text{SoilSat} + w3 * \tan(\text{Slope})$$

w1



w2



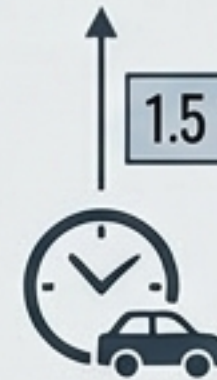
w3



GSI multi-variable hazard index combining real-time rain, soil shear saturation %, and topographic DEM slope angle (0-1 index).

Compute Engine

$$\text{Score} = \frac{\text{ICU_beds}}{(\text{Travel_time} + 1.5 * \text{Traffic_delay})}$$

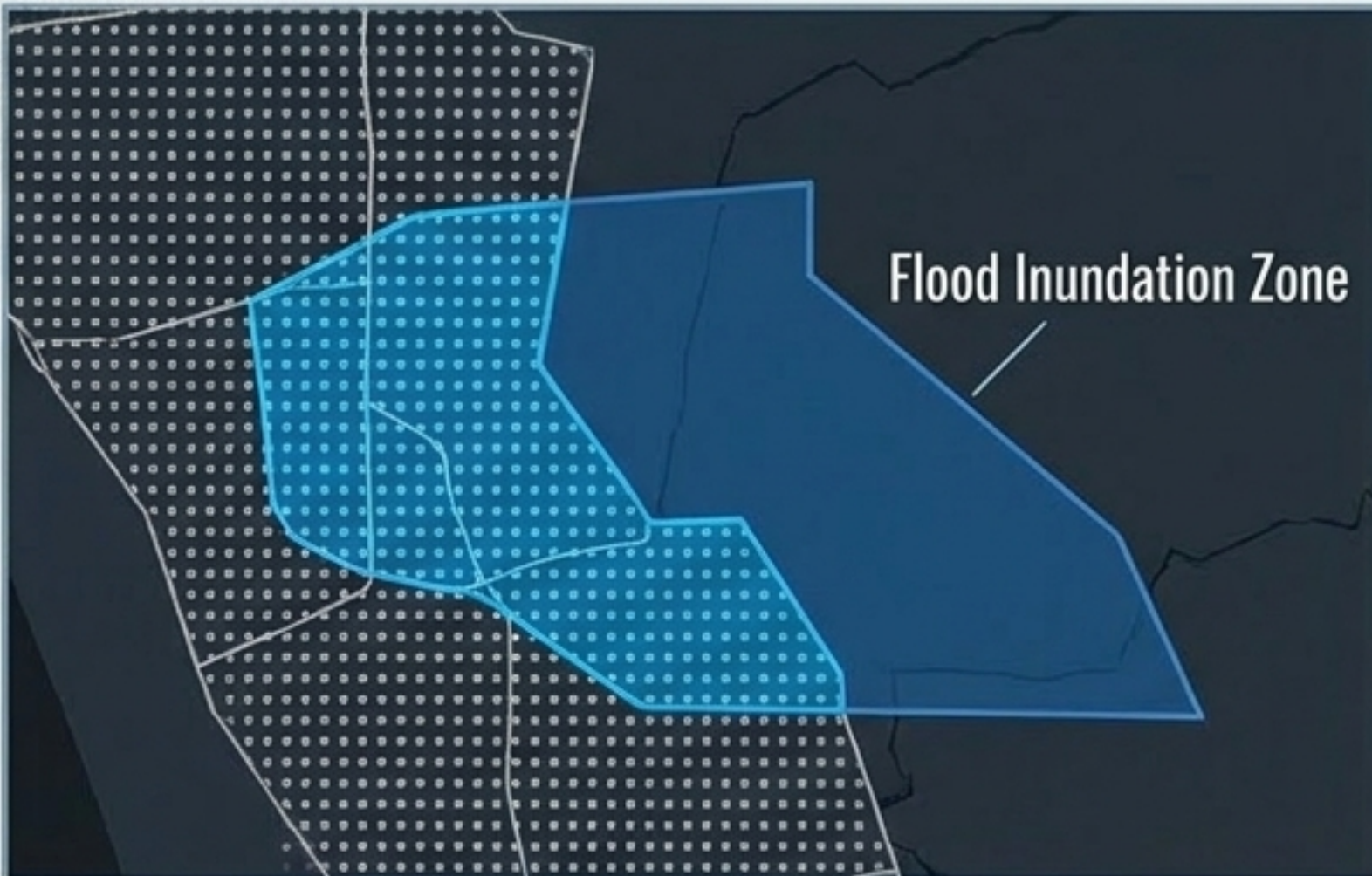


1.5 * Traffic Penalty



Ranks 108 trauma centers by real-time ICU bed availability penalized by MapmyIndia traffic ETAs.

Execution Engine: Spatial Targeting & Multilingual Action



Database spatial intersection overlaying CWC flood layers with Census demographics (ST_Intersects(census.geom, flood.geom)).
840+ Households Identified.

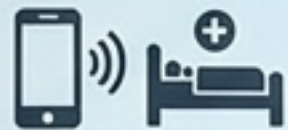
```
<alert
xmlns="urn:oasis:names:tc:emergency:cap:1.2">
  <language>ml-IN</language>
  <headline>വയനാട് മിന്നൽ പ്രളയ
മുന്നറിയിപ്പ്</headline>
  <area><circle>11.6854,76.1320
5000</circle></area>
</alert>
```

System autonomously generates OASIS CAP cell broadcast payloads to force Malayalam SMS alerts directly to specific cell towers within a 5000m radius.

The Synthesis: Telemetry to Execution Loop

A 4-stage continuous disaster monitoring and dispatch lifecycle.

04. EXECUTE (Action Dispatch)



Dispatches NDMA Sachet cell warnings & locks 108 ICU trauma beds.

03. SIMULATE (Plan Generation)

PostGIS and MapmyIndia API auto-calculate bypass evacuation routes.

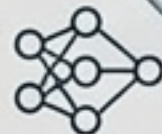


01. INGEST (Telemetry Streams)



Scrapes IMD doppler radar & CWC gauges via MCP every 15 minutes.

02. PROFILE (Risk Assessment)



LangGraph agents compute landslide susceptibility & downstream inundation zones.



The USP: Beyond Static Dashboards



Multi-Agency Data Fusion

Agents from IMD, CWC, NDRF, and 108 share a real-time semantic memory space.



CWC-to-Road Graph Intelligence

Flood inundation shapes directly overlay NHAI road graphs to auto-compute rescue detours.



Zero-Latency Hospital Triage

Queries health APIs to route casualties only to hospitals with verified ICU beds.



Multilingual Cell Broadcast

Pushes Hindi, English, and local languages directly to hazard zone cell towers.



Tested on Real Wayanad Data

Prototype modeled against Kabini reservoir discharge records and 2024 Wayanad flood maps.



100% India-Native Infrastructure

Zero dependency on foreign mapping or cloud tools. Built purely on Indian government APIs.

Projected Impact & National Expansion Roadmap

Performance Targets



< 5 min

Response time from CWC gauge trigger to NDMA cell broadcast (90% Warning Speedup).



75% Efficiency

Eliminates duplicate NDRF dispatches and optimizes ambulance routing.



85% Compliance

Localized multilingual broadcasts drastically increase citizen evacuation compliance.

Development Blueprint

Phase 3 (2027)

National scale covering all 36 States & UTs with cyclone, earthquake, and industrial modules.



Phase 2 (Q3 2026)

State-level deployment across Kerala & Uttarakhand SDMA. Live 108 Health API integrations.



Phase 1 (MVP/Now)

Core 9-agent architecture, FastMCP servers, Wayanad flood scenario POC.



BUILT FOR INDIA. BY INDIA.